

How Much Do Earth-Observation Foundation Models Help Flood Mapping When Labels Are Scarce?

Frozen, LoRA and Full Fine-Tuning of Prithvi-EO 2.0 on Sen1Floods11

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Abstract—Earth-observation foundation models are expected to reduce the number of labels needed for tasks such as flood mapping. We test this expectation in a controlled setting on the Sen1Floods11 benchmark. Two U-Nets (trained from scratch and ImageNet-initialized) are compared with Prithvi-EO 2.0, a 300M-parameter masked-autoencoder foundation model, under three fine-tuning regimes: frozen encoder, low-rank adaptation (LoRA) and full fine-tuning. All models see the same six Sentinel-2 bands and the same budget of optimizer steps, with 1 % to 100 % of the training labels (3 to 252 chips), three seeds per setting and 84 runs in total. The U-Nets lead at every label fraction: 0.829 versus 0.767 water IoU for Prithvi with LoRA at full labels, and 0.809 versus 0.704 with only three labeled chips. The gap therefore does not close as labels become scarce; it widens. Among the fine-tuning regimes, LoRA matches full fine-tuning (0.767 versus 0.770) with about 140 times fewer trainable parameters, and a frozen encoder trails both by about seven points. Pretraining also does not reduce the accuracy drop on the held-out Bolivia region. A pixel-level error analysis locates the foundation model’s shortfall at water edges and on water bodies smaller than 0.1 km², consistent with its 16-pixel patch resolution. For Sentinel-2 flood mapping on this benchmark, a small well-tuned CNN remains the better choice, even with a handful of labeled scenes.

Index Terms—flood mapping, Sentinel-2, foundation models, parameter-efficient fine-tuning, LoRA, Sen1Floods11, domain shift

I. INTRODUCTION

Rapid flood-extent mapping from optical satellite imagery is limited less by model capacity than by the cost of pixel-level labels: Sen1Floods11 [1], the standard benchmark, contains only 446 hand-labeled chips. Earth-observation (EO) foundation models such as Prithvi-EO 2.0 [2], pretrained by masked autoencoding on large unlabeled archives, promise to reduce that dependence on labels. Two questions matter to a practitioner who has a handful of labeled scenes and a single GPU:

- 1) How much accuracy does the foundation model buy over a well-tuned U-Net, and does the advantage grow as labels shrink?
- 2) Which fine-tuning regime—a frozen encoder with a trained head, low-rank adapters (LoRA) [3], or full fine-tuning—gives the best accuracy per trainable parameter, and which generalizes best to an unseen region?

We answer both with a fixed experimental matrix. Four models (two U-Nets, frozen and LoRA Prithvi) are trained at six label fractions from 100 % down to 1 % of the training chips, with three seeds each. Full fine-tuning and a set of ablations are added at full labels. Every run uses the same optimizer-step budget and is evaluated on the in-distribution test split and on the held-out Bolivia region of Sen1Floods11.

Contributions. (i) A controlled comparison of from-scratch and ImageNet-initialized U-Nets against frozen, LoRA and fully fine-tuned Prithvi-EO 2.0 on Sen1Floods11, down to three labeled chips, under a matched step budget. (ii) Evidence that on this benchmark the foundation model’s disadvantage grows rather than shrinks as labels become scarce, and that pretraining does not reduce the drop on the held-out region. (iii) A pixel-level error analysis that locates the foundation model’s excess errors at water edges and small water bodies. (iv) An open, reproducible pipeline (<https://github.com/mertsemih/eo-foundation-flood>) with every run’s configuration and metrics.

II. RELATED WORK

Flood mapping on Sen1Floods11. Bonafilia et al. [1] released Sen1Floods11 with co-registered Sentinel-1 and Sentinel-2 chips over eleven flood events, a hand-labeled subset and a held-out Bolivia event for testing transfer. Its fixed splits and small size have made it a common downstream task for EO foundation models.

EO foundation models. Prithvi [4] and Prithvi-EO 2.0 [2] pretrain vision transformers [5] as masked autoencoders [6] on Harmonized Landsat–Sentinel (HLS) imagery. Related models include SatMAE [7] and the sensor-agnostic DOFA [8]. Benchmarks such as GEO-Bench [9] and PANGAEA [10] evaluate such models across tasks. PANGAEA is the closest to our study: it reports that foundation models “do not consistently outperform supervised models”. On Sen1Floods11, its supervised U-Net baseline scores above Prithvi at 100 %, 50 % and 10 % of the labels (mIoU 91.42 vs. 90.37 at 100 %, 88.55 vs. 86.28 at 10 %). PANGAEA trains a decoder on frozen encoders for a fixed number of epochs and stops at 10 % of the labels. We extend that picture in three ways: we compare

frozen, LoRA and full fine-tuning; we hold the optimizer-step budget fixed across label fractions; and we go down to 1 % of the labels.

Parameter-efficient fine-tuning. Adapters [11] and LoRA [3] train a small number of added parameters while the pretrained weights stay fixed. Kumar et al. [12] showed that full fine-tuning can distort pretrained features and underperform linear probing out of distribution. This motivates our comparison of the three regimes on the held-out Bolivia region.

III. DATA

A. Sen1Floods11

We use the hand-labeled subset of Sen1Floods11: 446 chips of 512×512 pixels at 10 m over eleven flood events, each with Sentinel-2 L1C (13 bands), Sentinel-1 GRD (VV, VH) and a per-pixel label in $\{-1, 0, 1\}$ for no-data/cloud, no water and water. We follow the official split: 252 training, 89 validation and 90 test chips drawn from ten regions, plus 15 chips from Bolivia that appear in no other split and serve as the out-of-distribution (OOD) test.

Two properties of the data shape the evaluation. First, water is rare: the median chip has 2.5 % water among valid pixels (mean 9 % in training), so we report water IoU rather than pixel accuracy. Second, the Bolivia split is not only a new region but also a cloudier one: 27 % of its pixels are no-data on average against 13 % in the other splits. Any OOD claim therefore mixes geographic and cloud-cover shift; we return to this in Sec. VI.

B. Inputs and preprocessing

All models receive the same six Sentinel-2 bands (B2, B3, B4, B8A, B11, B12), which correspond to the six HLS bands Prithvi-EO was pretrained on. Restricting the U-Nets to the same bands isolates the effect of pretraining from that of input information. Reflectances are standardized per band with statistics of the training split. Training uses random 224×224 crops with flips and 90° rotations; evaluation uses whole 512×512 chips. Pixels labeled -1 are excluded from the loss and from all metrics.

C. Label-fraction protocol

For a fraction $f \in \{1, 0.25, 0.10, 0.05, 0.02, 0.01\}$ we keep $\text{round}(252f)$ training chips, i.e. 252, 63, 25, 13, 5 and 3 chips. The subset is drawn with a random permutation seeded by the run seed, so the three seeds vary both the initialization and the subset, and every model sees the same subset for a given seed. Validation and test sets are never subsampled. With a fixed number of epochs, low-label runs would get proportionally fewer parameter updates. We therefore train every run for the same budget of about 1,550 optimizer steps with the same schedule. Because whole epochs are run, the exact count is 1,550–1,575 steps: 50 epochs at $f = 1$, or 25 for full fine-tuning with its smaller batch. At $f \leq 0.02$ the batch size is reduced to the number of training chips.

IV. METHODS

A. Baselines

U-Net (scratch) is a U-Net [13] with a ResNet-34 [14] encoder (24.4 M parameters) trained from random initialization. **U-Net (ImageNet)** is the same architecture with an ImageNet-pretrained encoder, a control for “generic” versus “EO-specific” pretraining.

B. Prithvi-EO 2.0 with three fine-tuning regimes

Prithvi-EO 2.0 (300M) is a ViT-L masked autoencoder (24 blocks, embedding size 1024, patch size 16) pretrained on HLS imagery. We take the encoder, concatenate the token maps of its last four blocks and decode them with a light fully convolutional head (four upsampling stages, 1.4 M parameters). Three regimes differ only in which encoder parameters receive gradients:

- **Frozen:** encoder fixed, head only (1.4 M trainable).
- **LoRA:** rank-8 adapters ($\alpha = 16$, dropout 0.05) on the query/key/value projections of every block (0.8 M adapter parameters, 2.2 M trainable in total).
- **Full:** all 305 M parameters, encoder at a $20\times$ lower learning rate than the head.

C. Training and evaluation

All models use AdamW [15], a cosine learning-rate schedule with linear warm-up, gradient clipping at 1.0, mixed precision and an unweighted cross-entropy loss. The head (or the whole U-Net) uses a learning rate of 10^{-3} , except 5×10^{-4} for the ImageNet U-Net. The adapters use $0.5\times$ the head rate for LoRA, and the encoder $0.05\times$ for full fine-tuning. Weight decay is 0.01 (0.05 for full fine-tuning), warm-up is 100 steps (200 for full fine-tuning) and the batch size is 8. Full fine-tuning uses a batch of 4 to fit in 8 GB of GPU memory. Each run is validated about 50 times (25 times for full fine-tuning), and the checkpoint with the best validation water IoU is evaluated on the test and Bolivia splits. Water IoU is computed from the confusion matrix accumulated over all valid pixels of a split. We report the mean $\pm$ standard deviation over three seeds. Seed-0 runs used a single RTX 3070 Ti laptop GPU (8 GB); some later seeds ran on a Colab T4. The whole study took 12.7 GPU-hours.

V. RESULTS

A. Full-label regime

Table I reports the five main models at $f = 1$. Both U-Nets reach 0.82–0.83 test water IoU. The best Prithvi variants reach 0.767 (LoRA) and 0.770 (full fine-tuning), about six points lower, and the frozen encoder reaches 0.701. Every gap between a U-Net and a Prithvi variant is several times larger than the seed standard deviations, which are at most 0.015. ImageNet initialization adds 0.6 points over training from scratch, which is within the seed spread.

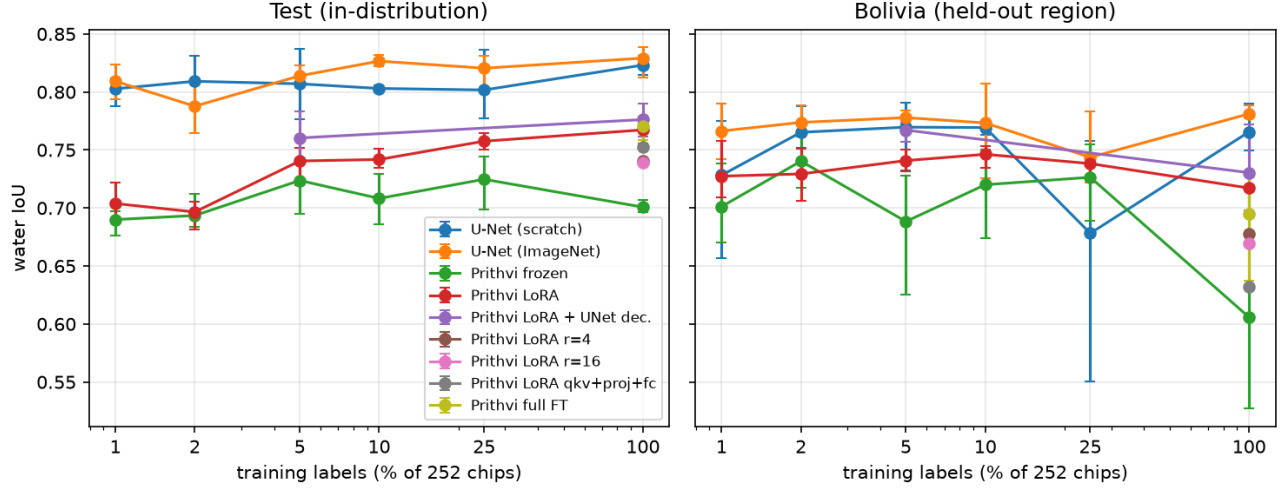


Fig. 1. Water IoU vs. training-label fraction (log scale; 100% = 252 chips) on the test split (left) and the held-out Bolivia region (right). Markers show the mean over three seeds, error bars the range (min-max). Points without a line are configurations run only at $f = 1$ (full fine-tuning and the LoRA rank/target ablations); the multi-scale decoder was run at 5% and 100%.

TABLE I

ALL TRAINING LABELS ($f = 1$). WATER IOU ON THE TEST SPLIT AND ON THE HELD-OUT BOLIVIA SPLIT, MEAN $\pm$ STD OVER THREE SEEDS. GPU MINUTES ARE FOR THE SEED-0 RUN ON AN RTX 3070 Ti (8 GB).

Model	Pretraining	Trainable	Test IoU _w	Bolivia IoU _w	GPU min
U-Net (scratch)	–	24.4 M	0.823 $\pm$ 0.008	0.766 $\pm$ 0.021	3.9
U-Net (ImageNet)	ImageNet	24.4 M	0.829 $\pm$ 0.015	0.781 $\pm$ 0.007	3.7
Prithvi frozen	Prithvi-EQ 2.0	1.4 M	0.701 $\pm$ 0.005	0.606 $\pm$ 0.083	8.7
Prithvi LoRA (r=8)	Prithvi-EQ 2.0	2.2 M	0.767 $\pm$ 0.011	0.717 $\pm$ 0.019	13.3
Prithvi full FT	Prithvi-EQ 2.0	305 M	0.770 $\pm$ 0.010	0.695 $\pm$ 0.050	9.0

TABLE II

TEST WATER IOU VS. TRAINING-LABEL FRACTION (NUMBER OF CHIPS IN PARENTHESES), MEAN $\pm$ STD OVER THREE SEEDS. BEST PER ROW IN BOLD.

Labels	U-Net (scratch)	U-Net (ImageNet)	Prithvi frozen	Prithvi LoRA (r=8)
100% (252)	0.823 $\pm$ 0.008	0.829 $\pm$ 0.015	0.701 $\pm$ 0.005	0.767 $\pm$ 0.011
25% (63)	0.802 $\pm$ 0.031	0.820 $\pm$ 0.016	0.725 $\pm$ 0.024	0.758 $\pm$ 0.007
10% (25)	0.803 $\pm$ 0.003	0.827 $\pm$ 0.005	0.708 $\pm$ 0.022	0.742 $\pm$ 0.009
5% (13)	0.807 $\pm$ 0.031	0.814 $\pm$ 0.010	0.724 $\pm$ 0.034	0.741 $\pm$ 0.018
2% (5)	0.809 $\pm$ 0.022	0.788 $\pm$ 0.021	0.694 $\pm$ 0.016	0.697 $\pm$ 0.013
1% (3)	0.803 $\pm$ 0.013	0.809 $\pm$ 0.015	0.690 $\pm$ 0.012	0.704 $\pm$ 0.017

B. Low-label regime

Fig. 1 and Table II show the label sweep. On the test split, no U-Net curve crosses a Prithvi curve. From 100% to 1% of the labels, both U-Nets lose about two points (0.829 $\rightarrow$ 0.809 and 0.823 $\rightarrow$ 0.803). With three labeled chips they are still above full-label LoRA. Prithvi with LoRA loses six points over the same range (0.767 $\rightarrow$ 0.704). The frozen encoder stays flat between 0.69 and 0.73, as expected for a model whose features do not adapt. The gap between the better U-Net and LoRA is 6.2 points at 100%, 7.3 at 5%, 11.3 at 2% and 10.5 at 1%. It is not monotone, but it grows from about six to about eleven points as labels become scarce. Every U-Net mean in Table II is at least 0.788, and every Prithvi mean is at most 0.767.

At 1–2% the seed also changes the subset. The spread

TABLE III

BOLIVIA (HELD-OUT REGION) WATER IOU VS. TRAINING-LABEL FRACTION.

Labels	U-Net (scratch)	U-Net (ImageNet)	Prithvi frozen	Prithvi LoRA (r=8)
100% (252)	0.766 $\pm$ 0.021	0.781 $\pm$ 0.007	0.606 $\pm$ 0.083	0.717 $\pm$ 0.019
25% (63)	0.678 $\pm$ 0.112	0.744 $\pm$ 0.035	0.726 $\pm$ 0.034	0.738 $\pm$ 0.010
10% (25)	0.769 $\pm$ 0.006	0.773 $\pm$ 0.043	0.720 $\pm$ 0.041	0.746 $\pm$ 0.010
5% (13)	0.770 $\pm$ 0.032	0.778 $\pm$ 0.008	0.688 $\pm$ 0.055	0.741 $\pm$ 0.009
2% (5)	0.765 $\pm$ 0.020	0.774 $\pm$ 0.014	0.740 $\pm$ 0.022	0.729 $\pm$ 0.023
1% (3)	0.728 $\pm$ 0.063	0.766 $\pm$ 0.024	0.701 $\pm$ 0.034	0.727 $\pm$ 0.027

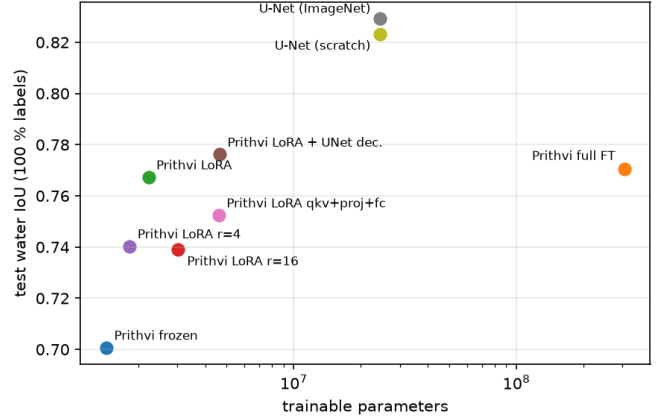


Fig. 2. Test water IoU at $f = 1$ vs. number of trainable parameters (log scale).

across seeds (up to 0.022 for the U-Nets) therefore includes the effect of which chips are drawn, and it stays well below the gap between model families.

C. Accuracy per trainable parameter

Fig. 2 places each configuration by trainable parameters. LoRA trains 2.2 M parameters, 0.7% of the 305 M updated by full fine-tuning, and reaches the same test accuracy (0.767

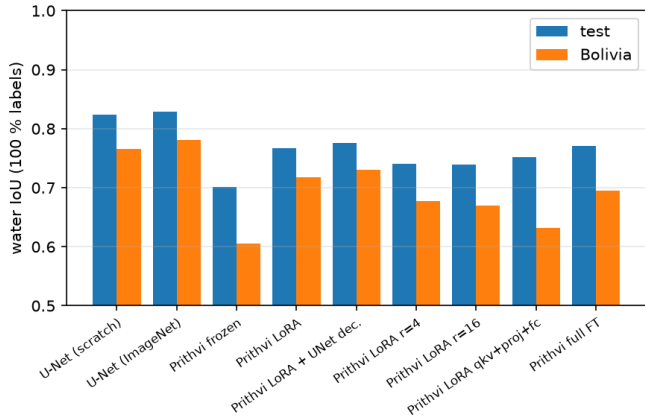


Fig. 3. Test vs. Bolivia water IoU at $f = 1$ for every configuration.

TABLE IV
ABLATIONS AT $f = 1$: LoRA RANK, LoRA TARGET MODULES, AND DECODER
($\dagger$: SINGLE SEED).

Variant	Trainable	Test IoU _w	Bolivia IoU _w
LoRA $r=4$, qkv	1.83 M	0.740 $\dagger$	0.678 $\dagger$
LoRA $r=8$, qkv (main)	2.23 M	0.767 $\pm$ 0.011	0.717 $\pm$ 0.019
LoRA $r=16$, qkv	3.01 M	0.739 $\dagger$	0.670 $\dagger$
LoRA $r=8$, qkv+proj+fc1+fc2	4.61 M	0.752 $\dagger$	0.632 $\dagger$
LoRA $r=8$, qkv + multi-scale U-Net decoder	4.65 M	0.776 $\pm$ 0.014	0.730 $\pm$ 0.038

± 0.011 vs. 0.770 ± 0.010), i.e. 99.6% of it. Adding 0.8 M adapter parameters to the frozen encoder is worth 6.7 points. Beyond that, neither more adapter capacity nor about $140\times$ more trainable parameters improves Prithvi, and neither closes the gap to the 24 M-parameter U-Nets.

D. Transfer to Bolivia

All models lose accuracy on Bolivia (Fig. 3). At $f = 1$ the drop from test to Bolivia is 4.8 points for the ImageNet U-Net, 5.8 for the scratch U-Net, 5.0 for LoRA, 7.5 for full fine-tuning and 9.5 for the frozen encoder. EO pretraining therefore does not shrink the transfer gap. The ImageNet U-Net has the highest Bolivia accuracy at every label fraction (Table III), and among the five main models it has the smallest drop. LoRA’s Bolivia score (0.717 ± 0.019) is above that of full fine-tuning (0.695 ± 0.050). This is consistent with the observation of Kumar et al. [12], but the difference lies inside the spread of full fine-tuning and is not conclusive.

Bolivia results are also noisier than test results. The scratch U-Net at 25% has one seed at 0.55 (std 0.112), and LoRA’s Bolivia score is not monotone in the label fraction: 0.746 at 10% is higher than 0.717 at 100%. With 15 chips and about a quarter of their pixels missing, the split ranks model families reliably but not neighbouring configurations.

E. Ablation

Table IV varies the LoRA configuration. Among ranks 4, 8 and 16, rank 8 on the query/key/value projections gave the best seed-0 result (0.740, 0.778 and 0.739; 0.767 ± 0.011 over three seeds for rank 8). Adding adapters to the attention output

and MLP layers doubles the trainable parameters and lowers the Bolivia score to 0.632. These single-seed differences of three to four points are suggestive only.

To test whether the light head limits Prithvi, we replaced it with a multi-scale U-Net-style decoder over the outputs of blocks 6, 12, 18 and 24 (4.65 M trainable parameters). Over three seeds this decoder reaches 0.776 ± 0.014 at $f = 1$, equal to the FCN head within noise, and 0.760 ± 0.022 at 5% (FCN head: 0.741 ± 0.018). A stronger decoder helps slightly when labels are scarce, but it leaves a gap of about five points to the U-Nets. The shortfall therefore lies in the encoder representation for this task, not in the head.

F. Where the errors are

To see what the aggregate numbers hide, we predicted every test and Bolivia chip with the seed-0 checkpoints at $f = 1$ and analysed the errors pixel by pixel. Fig. 4 shows four chips selected by a fixed rule. Two patterns are visible. First, Prithvi misses narrow channels and small ponds that the U-Net recovers, such as the river along the bottom of the Spanish chip. Second, its water outlines are smoother than the label, with bands of false and missed water along the edges. On Bolivia, the Prithvi variants miss much of the water (recall 0.53–0.77 over the split, versus 0.83–0.89 for the U-Nets), while the U-Net adds some false water.

Two measurements quantify these patterns on the test split. They are robust to the thresholds we tried. *Water-body size*: we split the labeled water into connected components. Water bodies smaller than 1,000 pixels (0.1 km^2) hold 11% of the water pixels. On these, recall is 0.68–0.69 for the U-Nets and 0.48 (LoRA) and 0.43 (full) for Prithvi. On larger bodies the U-Nets and the fine-tuned Prithvi variants all reach 0.90–0.93. The same ordering holds at thresholds of 500 and 2,000 pixels. *Edges*: 77% of Prithvi’s errors lie within 3 pixels of a label water edge, against 64–65% for the U-Nets. In absolute terms, LoRA makes 484k error pixels near edges versus 319k–342k for the U-Nets, and fewer errors far from edges (144k versus 172k–192k). Prithvi’s entire excess error is thus concentrated at water boundaries. Within 1 pixel of the edge, the error shares are equal for all models (44–46%); the difference appears 2–5 pixels away from the edge. Prithvi therefore places boundaries a few pixels off, rather than failing on the edge pixels themselves. Its precision is on par with the U-Nets (0.89–0.91); the lost accuracy is recall (0.85–0.86 versus 0.89–0.90).

VI. DISCUSSION

Claims revisited. Before running the experiments we expected (1) at most a marginal advantage for the foundation model with full labels, (2) a widening advantage at 5–10% labels, (3) LoRA recovering most of full fine-tuning at a fraction of its parameters, and (4) pretraining helping more on Bolivia than in distribution. Claim (1) holds in the negative direction: the U-Nets are ahead by six points. Claim (2) is refuted down to 1%; the gap is largest at 1–2% (10.5–11.3 points). Claim (3) holds, and LoRA matches full fine-tuning

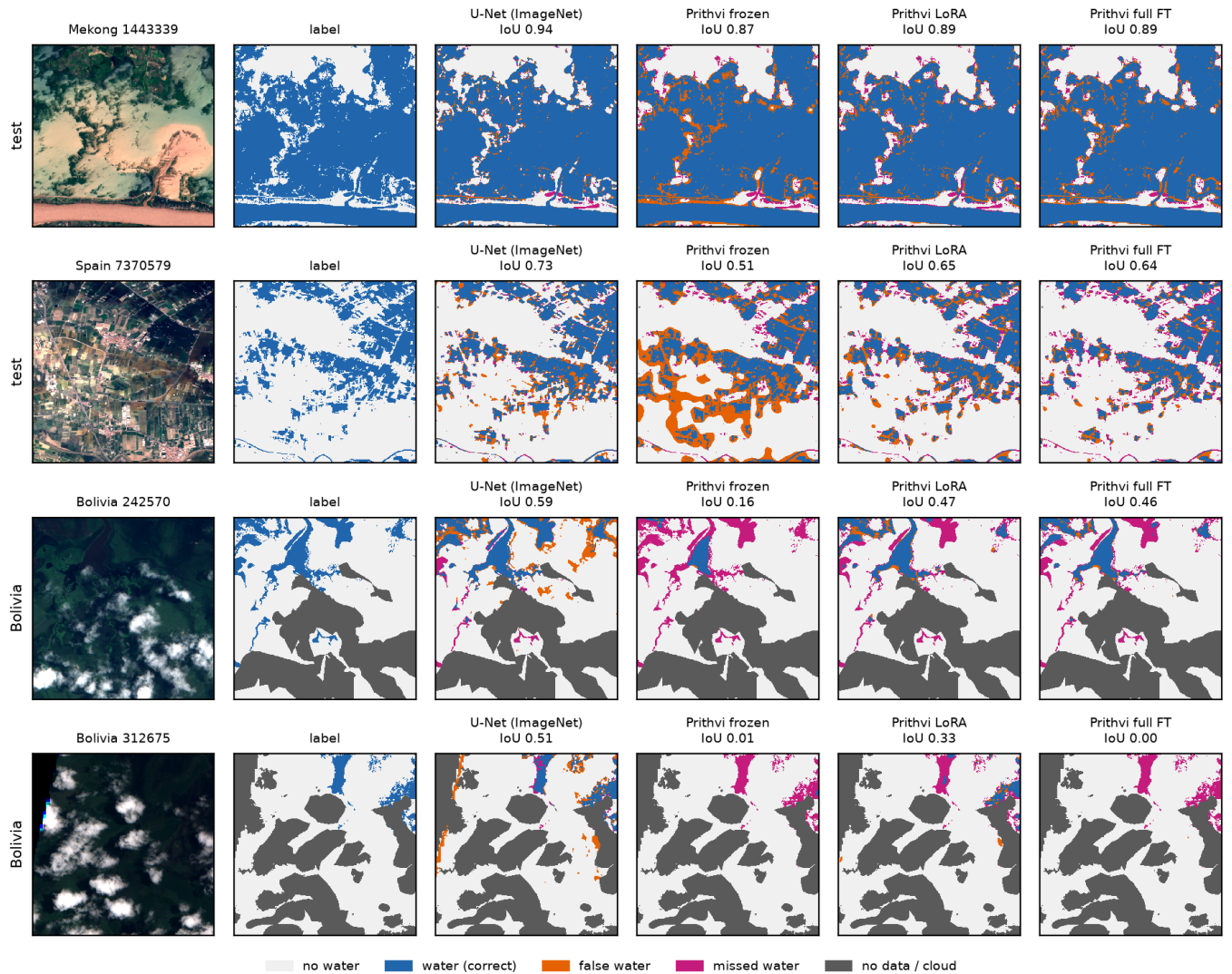


Fig. 4. Predictions of the seed-0 models trained with all labels, drawn as error maps against the hand label. Rows are chosen by a fixed rule rather than by eye: among chips with at least 5 % water, the chips at the 50th and 75th percentile of the per-chip IoU difference between the ImageNet U-Net and Prithvi LoRA, for the test split (top) and Bolivia (bottom). Titles give the per-chip water IoU.

outright. Claim (4) is refuted: the ImageNet U-Net transfers best among the main models.

Why does the U-Net keep up with three chips? Segmentation labels are dense. A 512×512 chip provides up to 262k labeled pixels, and three chips can hold on the order of 10^5 water pixels. For a convolutional network with local receptive fields and full-resolution skip connections, that is a substantial training signal, especially when the step budget is held constant. Our results agree with PANGAEA [10], which found the U-Net ahead of Prithvi on Sen1Floods11 down to 10 % labels under a different protocol.

Why does Prithvi fall behind? The error analysis points to spatial resolution. The ViT sees the image as 16×16 patches, so its features have $1/16$ of the input resolution. Its excess errors are exactly where that should hurt: along water edges and on water bodies below 0.1 km^2 . On large water bodies

it is as accurate as the U-Net, and it makes fewer errors far from edges. The multi-scale decoder does not remove the gap, since all of its inputs come from the same $1/16$ -resolution token grid; a decoder with a skip connection from the full-resolution input would be the natural next test. A second, untested factor is the input domain. Prithvi was pretrained on 30 m HLS surface reflectance, while Sen1Floods11 provides 10 m Sentinel-2 top-of-atmosphere (L1C) reflectance.

The Bolivia confound. Because the Bolivia chips are both from a new region and cloudier, the drop we measure mixes geographic shift and missing data. Separating the two would require a cloud-matched OOD split. Its small size also limits how finely models can be ranked. We therefore state only that no EO-pretrained model transfers clearly better than the ImageNet U-Net.

Practical recommendation. For a practitioner with a few

labeled Sentinel-2 flood scenes and one consumer GPU, an ImageNet-initialized U-Net is the most accurate option we tested. It is also the cheapest, at about 4 GPU-minutes per run versus 9–13 for Prithvi. If a foundation model is required, for example for consistency with other tasks, LoRA gives full fine-tuning accuracy with under 1 % of the trainable parameters and checkpoints of a few megabytes.

VII. LIMITATIONS

The study covers a single dataset and sensor (Sentinel-2 only, no SAR fusion) and a single foundation model at one size. We did not tune hyperparameters per label fraction, and the consumer-GPU budget limits full fine-tuning to a batch of 4, so within the same step budget it sees half as many training samples as the other models. The Bolivia OOD split is small (15 chips) and cloudier than the rest. The LoRA rank and target ablations and the pixel-level error analysis use one seed. Absolute numbers are not comparable to those reported for Prithvi elsewhere [2], which use different band sets, decoders and training budgets; our claims concern the relative ordering under a matched protocol.

VIII. CONCLUSION

On Sen1Floods11 with Sentinel-2 input and a matched training budget, U-Nets outperform Prithvi-EO 2.0 by six points with full labels and by ten points with three labeled chips. The foundation model’s advantage in the low-label regime did not materialize, and pretraining did not reduce the loss of accuracy on a held-out region. Its excess errors lie at water edges and on small water bodies, which points to the 16-pixel patch resolution as the limiting factor. Within the foundation-model family, LoRA is the regime of choice: it matches full fine-tuning with about 1/140 of the trainable parameters.

REPRODUCIBILITY

Code, configurations and per-run metrics are available at <https://github.com/mertsemih/eo-foundation-flood>. Each run is fully specified by one YAML file; the dataset is downloaded with one script from the public Sen1Floods11 bucket.

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